Forecasting, Part 3: Regression and Seasonality

Chapter 16, §16.3–16.4

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Objectives

- Forecast with **trend projection** — a regression of the series on the period number — and compute its MAD.
- Test the residuals for **autocorrelation** with the **Durbin–Watson** statistic, and say what a positive result does to your regression.
- Forecast with **multiplicative decomposition**: seasonal factors, a deseasonalised trend, and their product.
- Represent seasonality with **dummy variables** in a multiple regression.
- Choose among all of Chapter 16's methods for each challenge series, and defend the choice with a backtest.

> You now have Chapters 14 and 15. That changes what you can do with a time series: instead of averaging its past, you can *model* it — put the period number on the right-hand side and let regression find the trend; put the season on the right-hand side and let it find the calendar. The methods before Exam 1 assumed a level. The methods before Exam 2 added a drifting level. These add a trend and a season, which is everything in Donnelly's list of components except the cycle.

The smoothing methods were atheoretic: they had nothing to say about *why* a series moves. Regression is at least a hypothesis — sales grow with time; March is bigger than February — and a hypothesis can be wrong in ways you can check. That is the point of the Durbin–Watson test below.

Trend Projection

Trend projection fits a straight line to the series against the period number and extends it into the future (Formula 16.6):

$$\hat{y}_t = b_0 + b_1 t,$$

with b_1 and b_0 the ordinary least-squares slope and intercept from Chapter 14 (Formulas 16.7 and 16.8), $t = 1, 2, \ldots, n$.

For Brandywine Ford, t runs 1 to 12 and the regression is simple enough to do by hand: $\sum t = 78$, $\sum y = 1,599$, $\sum ty = 10,783$, $\sum t^2 = 650$.

$$\hat{y}_t = 115.5455 + 2.7238\,t \qquad \hat{y}_{13} = 115.5455 + 2.7238\,(13) = 150.95 \approx 151 \ \text{cars}$$

Sales grow by about 2.72 cars a quarter. The forecast for period 13 is 151; for period 14 it would be 154, and each period further out is a little less believable — a line extended is a line trusted.

	t	actual	fitted	residual	abs_error
2016Q1	1	97.00	118.30	−21.30	21.30
2016Q2	2	142.00	121.00	21.00	21.00
2016Q3	3	108.00	123.70	−15.70	15.70
2016Q4	4	135.00	126.40	8.60	8.60
2017Q1	5	120.00	129.20	−9.20	9.20
2017Q2	6	164.00	131.90	32.10	32.10
2017Q3	7	126.00	134.60	−8.60	8.60
2017Q4	8	150.00	137.30	12.70	12.70
2018Q1	9	123.00	140.10	−17.10	17.10
2018Q2	10	151.00	142.80	8.20	8.20
2018Q3	11	141.00	145.50	−4.50	4.50
2018Q4	12	142.00	148.20	−6.20	6.20

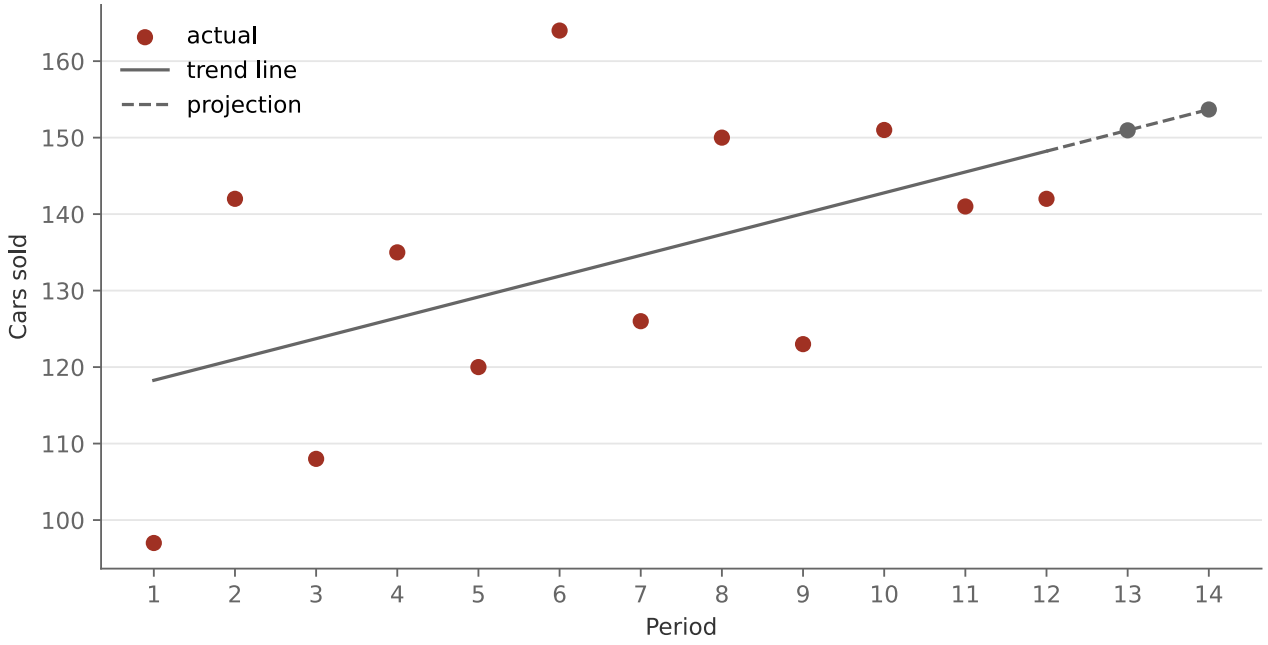
The forecasting error of a regression is its **residual**, $e_t = y_t - \hat{y}_t$ (Formula 16.9), and the MAD is the mean absolute residual: 13.8 cars. Note that this MAD averages all twelve residuals — regression forecasts every period, including the first.

	coef	std error	t	p-value
Intercept	115.55	10.63	10.87	NaN
Period	2.72	1.44	1.89	NaN

This is what Excel's Regression tool prints (Figure 16.11C): $R^2 = 0.2625$, standard error 17.27, and a p-value of nan on the slope. At $\alpha = 0.05$ the trend is *not* statistically significant — and yet its MAD is among the best so far.

Those are different questions. Significance asks whether the slope could be zero given twelve noisy quarters; the MAD asks how far off the forecasts were. A line can forecast usefully without proving anything.

Figure 16.9: forecasting with a trend projection



Checking for Autocorrelation

Regression assumes the residuals are independent of one another. In a time series that assumption fails more often than it holds, because whatever pushed this quarter's sales above the line is usually still pushing next quarter.

Autocorrelation is the condition in which the residuals of a regression are not independent of one another. The **Durbin–Watson statistic** tests for it (Formula 16.10):

$$d = \frac{\sum_{t=2}^{n} (e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2}.$$

d ranges from 0 to 4. A value near 2 means no autocorrelation; near 0, perfect positive autocorrelation (residuals in long runs of the same sign); near 4, perfect negative autocorrelation (residuals that alternate).

If consecutive residuals are close to each other, the numerator is small and d falls toward zero. Donnelly's Table 16.10 makes the point with five residuals all equal to 20: every difference is zero, $d = 0$, and knowing one residual tells you the next exactly. Negative autocorrelation is rare in business data, so the test is one-tailed:

$$H_0 : \text{no positive autocorrelation} \qquad H_1 : \text{positive autocorrelation is present}.$$

The test needs at least 15 observations, so the book switches to a 16-year series of union membership as a share of the labour force (Table 16.12):

	residual	prior	(e_t - e_t-1)^2	e_t^2
2002	0.0596	NaN	NaN	0.0035
2003	−0.1959	0.0596	0.0653	0.0384

	residual	prior	(e_t - e_t-1)^2	e_t^2
2004	−0.1513	−0.1959	0.0020	0.0229
2005	−0.4068	−0.1513	0.0653	0.1655
2006	−0.1622	−0.4068	0.0598	0.0263
2007	0.2824	−0.1622	0.1976	0.0797
2008	0.3269	0.2824	0.0020	0.1069
2009	0.5715	0.3269	0.0598	0.3266
2010	0.1160	0.5715	0.2074	0.0135
2011	0.1606	0.1160	0.0020	0.0258
2012	−0.1949	0.1606	0.1263	0.0380
2013	−0.0503	−0.1949	0.0209	0.0025
2014	−0.1057	−0.0503	0.0031	0.0112
2015	0.0388	−0.1057	0.0209	0.0015
2016	−0.2166	0.0388	0.0653	0.0469
2017	−0.0721	−0.2166	0.0209	0.0052

$$\hat{y}_t = 13.0850 - 0.1446\,t \qquad d = \frac{0.9185}{0.9143} = 1.0046$$

The critical values come from the Durbin–Watson table (Appendix A, Table 9), indexed by n and by k, the number of independent variables. For $n = 16, k = 1, \alpha = 0.05$: d_L = 1.1 and d_U = 1.37. The decision rule has three regions:

Result	Conclusion
$d < d_L$	Reject H_0: positive autocorrelation is present
$d_L \le d \le d_U$	Inconclusive
$d > d_U$	Fail to reject H_0: no evidence of positive autocorrelation

Here d = 1.005 $< d_L$, so we **reject H0: positive autocorrelation is present**. (The book gets 1.0085 from residuals rounded to three decimals; the difference is rounding, not disagreement.)

> What does a positive result cost you? The point estimates b_0 and b_1 are still unbiased; the *standard errors* are wrong, usually too small, so the t-test on the slope claims more certainty than it has. The trend line may still forecast tolerably. But autocorrelated residuals are also a message: there is information in the residuals a better model could use. If the last three residuals were negative, the next one probably is too — and the trend projection ignores that, while the naive forecast, without knowing it, uses it. The fix Donnelly suggests is to add an explanatory variable. Another, which you will meet if you take a time-series course, is to model the residuals themselves.

Forecasting with Seasonality

Look at the car sales again: quarters 2 and 4 spike, every year. Every method so far has treated that as noise. It is not noise; it is the calendar, and a calendar can be forecast.

Multiplicative Decomposition

The **multiplicative decomposition** model writes each observation as the product of its components (Formula 16.11):

$$y_t = T_t \times S_t \times R_t,$$

trend, seasonal and random. (The cycle is left out: three years of data are not enough to see one.) The forecast is $F_t = T_t \times S_t$ (Formula 16.14) — the random component is, by definition, unforecastable.

Four steps: identify the seasonal factors, deseasonalise the data, fit a trend to the deseasonalised data, and multiply.

Step 1: the centered moving average

Average four consecutive quarters. Because four is even, the average of periods 1–4 belongs to period 2.5, halfway between 2 and 3; so average the period-2.5 and period-3.5 averages to get a value centred *on* period 3. That is the **centered moving average**, and it contains one of every season, so the seasonal component has been averaged out of it — and so, roughly, has the random component. What is left is the trend: $\mathrm{CMA}_t \approx T_t$.

Step 2: the ratio to moving average

Divide each actual by its CMA (Formula 16.12): $\mathrm{RMA}_t = y_t / \mathrm{CMA}_t = S_t \times R_t$. The ratio is what the season and the noise did to that period.

	actual	CMA	RMA	season
2016Q1	97.0000	NaN	NaN	1
2016Q2	142.0000	NaN	NaN	2
2016Q3	108.0000	123.3750	0.8754	3
2016Q4	135.0000	129.0000	1.0465	4
2017Q1	120.0000	134.0000	0.8955	1
2017Q2	164.0000	138.1250	1.1873	2
2017Q3	126.0000	140.3750	0.8976	3
2017Q4	150.0000	139.1250	1.0782	4
2018Q1	123.0000	139.3750	0.8825	1
2018Q2	151.0000	140.2500	1.0766	2
2018Q3	141.0000	NaN	NaN	3
2018Q4	142.0000	NaN	NaN	4

Read period 3: CMA 123.375, actual 108, ratio 0.8754 — the third quarter of 2016 ran about 12% below trend. The first two and last two periods have no CMA: a centred average needs data on both sides.

Step 3: seasonal factors, then normalise them

Each quarter appears twice in the ratios. Average them, and the random component — which is what differs between the two years — cancels, leaving the seasonal factor. Then rescale so the four factors sum to exactly 4, because a "season" that averages more than 100% of an average quarter is not a season.

season	SF	NSF
1	0.8890	0.8958
2	1.1320	1.1406
3	0.8865	0.8932
4	1.0623	1.0704

The sum of the raw factors is 3.9698; multiplying each by 4 / 3.9698 gives the normalised factors. Quarter 2 runs about 14% above an average quarter; quarters 1 and 3 run about 10% below.

Step 4: deseasonalise, fit the trend, multiply

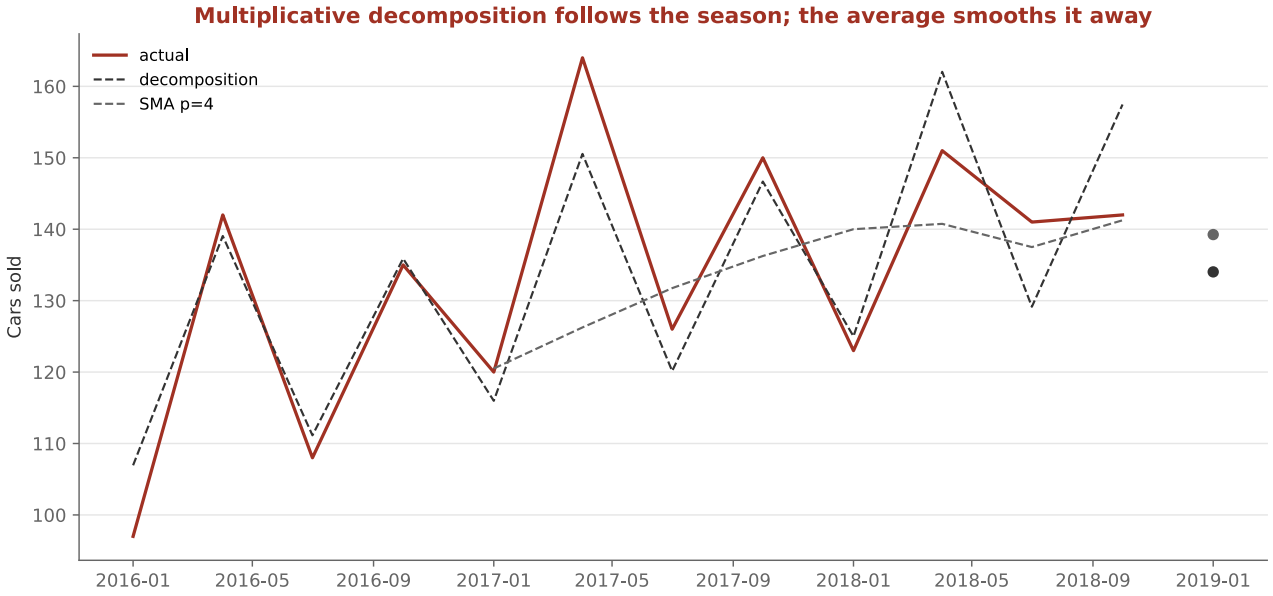
Divide each observation by its quarter's factor (Formula 16.13) and the seasonal spikes disappear. Regress what is left on the period number; that is the trend component. The forecast is the projected trend times the factor for the quarter being forecast.

	actual	season	S	deseasonalized	T	forecast	error	abs_error
2016Q1	97.00	1	0.90	108.29	119.39	106.95	−9.95	9.95
2016Q2	142.00	2	1.14	124.50	121.91	139.05	2.95	2.95
2016Q3	108.00	3	0.89	120.91	124.43	111.14	−3.14	3.14
2016Q4	135.00	4	1.07	126.12	126.95	135.89	−0.89	0.89
2017Q1	120.00	1	0.90	133.96	129.47	115.97	4.03	4.03
2017Q2	164.00	2	1.14	143.79	131.99	150.54	13.46	13.46
2017Q3	126.00	3	0.89	141.06	134.51	120.14	5.86	5.86
2017Q4	150.00	4	1.07	140.13	137.03	146.67	3.33	3.33
2018Q1	123.00	1	0.90	137.31	139.54	125.00	−2.00	2.00
2018Q2	151.00	2	1.14	132.39	142.06	162.04	−11.04	11.04
2018Q3	141.00	3	0.89	157.86	144.58	129.14	11.86	11.86
2018Q4	142.00	4	1.07	132.66	147.10	157.46	−15.46	15.46

$$T_t = 116.8744 + 2.5189\,t \qquad T_{13} = 149.62 \qquad F_{13} = T_{13} \times S_1 = 149.62 \times 0.8958 = 134.0 \approx 134 \ \text{cars}$$

	t	season	T	S	forecast
0	13	1	149.62	0.90	134.03
1	14	2	152.14	1.14	173.53
2	15	3	154.66	0.89	138.14
3	16	4	157.18	1.07	168.24

The trend alone would have said 150 cars; the first quarter's factor pulls it down to 134. And the MAD is 7.0 — less than half of any method before it. (The book's trend line, $116.85 + 2.52t$, was fitted to deseasonalised values rounded to one decimal; ours is fitted to the unrounded ones.)



Seasonality with Dummy Variables

Chapter 15 gave you dummy variables. A season is a qualitative variable with four categories, so it needs three dummies (one category is the base):

$$F_t = b_0 + b_1 t + b_2\,\mathrm{SD}_2 + b_3\,\mathrm{SD}_3 + b_4\,\mathrm{SD}_4,$$

where $\mathrm{SD}_q = 1$ in quarter q and 0 otherwise, and the first quarter is the base. Each b_q is how many more cars quarter q sells than the first quarter *at the same point on the trend*.

The book runs best-subsets regression and keeps the period and two of the three dummies — the third-quarter dummy is dropped, because the third quarter is not distinguishable from the first. Dummies are named here by quarter, so the book's SD_1 and SD_3 are our SD_2 and SD_4.

	coef	se	t	p
intercept	104.17	6.06	17.19	NaN
t	2.50	0.79	3.15	NaN
SD2	33.17	6.49	5.11	NaN
SD4	18.17	6.68	2.72	NaN

Adjusted R^2 is 0.771 and every coefficient has $p < 0.05$. The second quarter sells 33.2 more cars than the first; the fourth, 18.2 more. That is the decomposition's seasonal factor of 1.14 said in cars rather than ratios.

$$F_{13} = 104.1667 + 2.5\,(13) + 33.1667\,(0) + 18.1667\,(0) = 136.7 \approx 137 \ \ \text{cars}$$

	t	season	forecast
0	13	1	136.70
1	14	2	172.30
2	15	3	141.70
3	16	4	162.30

Table 16.22: every method, one table

method	next_forecast	MAD	RMSE	MAPE	n_errors
naive	142.00	26.64	29.70	19.72	11
SMA p=3	144.67	15.93	19.87	11.24	9
SMA p=4	139.25	11.16	16.02	7.60	8
WMA 3-2-1	143.17	15.28	19.24	10.81	9
Exp. smoothing 0.6	141.94	19.03	23.32	13.77	11
Exp. smoothing w/ trend 0.3, 0.1	148.37	16.53	21.81	11.67	11
Trend projection	150.95	13.76	15.76	10.58	12
Multiplicative decomposition	134.03	7.00	8.50	5.21	12
Seasonal dummies	136.67	6.50	7.50	4.87	12

The two methods that model *both* trend and season win by a wide margin, and the four-period moving average — the only smoothing method whose window spans every season — is next. Complexity is not what wins. Matching the method to the components that are actually in the series is what wins, and a method that models a component the series does not have loses ground for it. The next section shows both halves of that.

The Challenge Series

Las Vegas visitor volume: twelve seasons

Visitor volume is monthly, so there are twelve seasons, and the LVCVA does not seasonally adjust it. This is the challenge series with the strongest calendar in it.



2020 and 2021 are not a season; they are a pandemic. A seasonal factor for April averaged across a closed city and an open one describes neither. Every seasonal model below starts in January 2022. That is a judgment, and it is the kind of judgment the data cannot make for you.

season	1	2	3	4	5	6	7	8	9	10	11	12
NSF	0.990	0.927	1.055	1.010	1.045	0.990	1.018	0.975	0.980	1.060	0.968	0.983

The factors say October runs about 6% above an average month and February about 7% below. The deseasonalised trend is 3,336,649 − 668 visitors per month — read the sign; it is the post-2022 direction of the market, and it is what the seasonal factor gets multiplied by.

	t	season	T	S	forecast
0	56	8	3,299,244	0.975	3,216,156
1	57	9	3,298,576	0.980	3,231,782
2	58	10	3,297,908	1.060	3,494,571

	t	season	forecast
0	56	8	3,242,062
1	57	9	3,264,787
2	58	10	3,532,487

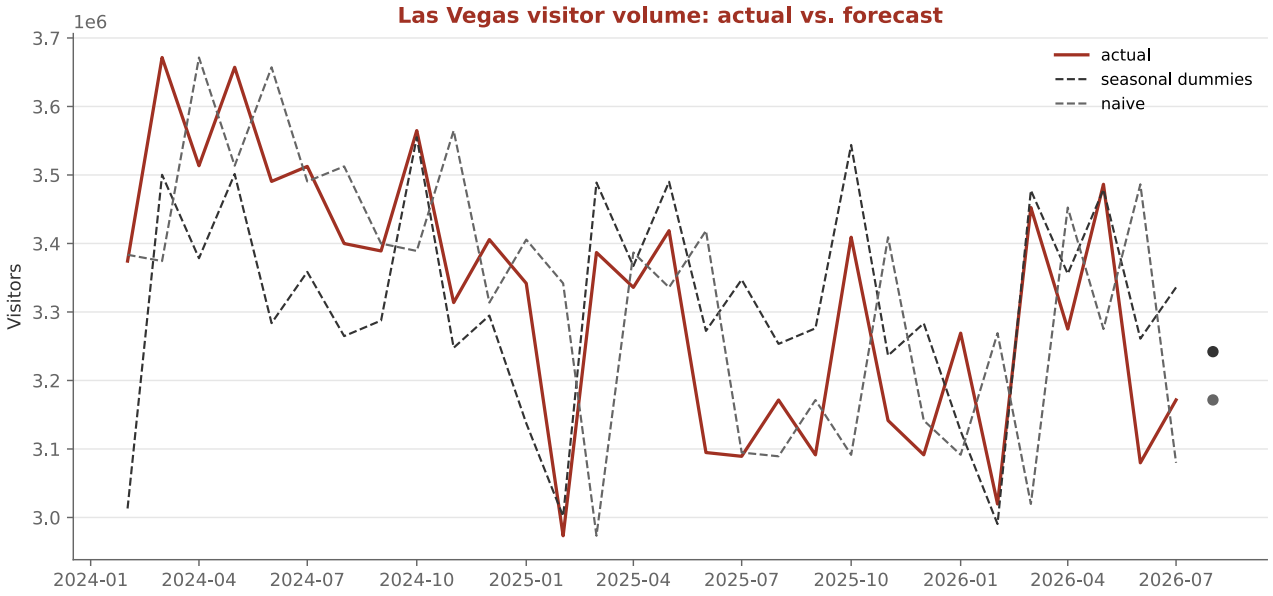
method	next_forecast	MAD	RMSE	MAPE	n_errors
naive	3,171,600	186,004	238,420	6	54
SMA p=3	3,245,933	153,872	193,898	5	52
SMA p=12	3,221,608	129,109	163,546	4	43
ES 0.3	3,225,583	167,522	229,777	5	54
Trend projection	3,306,511	160,327	225,445	5	55
Decomposition	3,216,156	121,541	176,211	4	55
Seasonal dummies	3,242,062	121,438	167,731	4	55

method	MAD	RMSE	MAPE	periods
Decomposition	133,268	155,854	4	12
SMA p=12	144,945	160,759	4	12
ES 0.3	148,969	171,384	5	12
Seasonal dummies	173,053	199,116	5	12
naive	212,042	244,796	7	12

Backtested over the last twelve months, the best method for 2026-08 is **Decomposition**, and the naive forecast — which knows nothing about calendars — is at 212,042. The twelve-month moving average does respectably for the same reason the four-quarter average did on car sales: its window contains every season once.

One more thing to check, now that you know how. The Durbin–Watson statistic on the dummy model's residuals is 0.488 against d_L = 1.53: **reject H0: positive autocorrelation is present**. The residuals run in streaks — a slow spring is followed by a slow summer — and the seasonal model cannot see that. That is exactly the situation in

which a forecast that blends the model with the latest residual will beat the model alone, and it is the next thing you would learn in a forecasting course.



Nevada house prices: a very significant line that keeps missing

date	2024Q3	2024Q4	2025Q1	2025Q2	2025Q3	2025Q4	2026Q1	2026Q2
residual	5.800	3.550	−3.130	−7.680	−18.300	−23.100	−27.810	−29.800

46 quarters from 2015, and the trend line is T_t = 244.7 + 9.41 t with R^2 = 0.967 and a slope p-value of nan. The forecast for 2026Q3 is 687.0; the naive forecast is 647.8. The Durbin–Watson statistic is 0.183: **reject H0: positive autocorrelation is present**.

Look at the last eight residuals in the table above: residuals that keep the same sign for quarters at a time are what a small d looks like. The line is straight and the index is not: it rose fast, then slowed, and a line fitted through both halves misses the recent quarters in the same direction every time. A regression can be overwhelmingly significant and still be the wrong shape. The naive forecast, which knows only where the index *is*, has the last residual built in; the trend-adjusted smoothing from Part 2 tracks the slope as it changes. Check both against the line in your backtest before you submit a trend projection for this series.

Housing starts: read the units line first

season	1	2	3	4	5	6	7	8	9	10	11	12
NSF	1.032	1.050	0.991	0.954	0.965	0.988	0.993	0.990	0.986	0.988	1.012	1.051

FRED's units line for HOUST reads *Thousands of Units, Seasonally Adjusted Annual Rate*. The Census Bureau removed the season before FRED published the number, and the factors a decomposition finds anyway range only 0.954 to 1.051. A seasonal model here is fitting noise, and it will cost you accuracy for the privilege. The same is true of the CPI and of initial claims (both SA); it is **not** true of the Nevada house price index (NSA) or of visitor volume.

method	MAD	RMSE	MAPE	periods
SMA p=3	76.3	99.4	5.8	12
ES 0.3	77.3	100.7	5.9	12

method	MAD	RMSE	MAPE	periods
SMA p=6	80.0	101.8	6.1	12
Trend	105.2	126.3	8.2	12
naive	107.6	132.8	8.1	12

The map of the challenge, after three notes. Each series has a method that suits it, and the reasons are the components in Donnelly's list. The CPI: trend, no season → trend-adjusted smoothing. Initial claims: noise around a level → a short moving average or exponential smoothing with a moderate α. Housing starts: already seasonally adjusted, noisy → a short moving average. The Nevada HPI: a trend that bends → smoothing that tracks the slope, not a fixed line. Visitor volume: a strong calendar → decomposition or dummies, refit as the months arrive. Whether that map is still right next month is what the backtest is for.

Working with Claude

The lab prompt

RUN THIS WITH CLAUDE

Get the MBA 775 course materials from https://github.com/jcrooker/mba775. I need `scripts/16c_regression_seasonality.py`, `scripts/_course.py`, `scripts/_forecast.py`, `scripts/_charts.py`, `data/lv_visitors.csv`, `data/nvsthpi.csv` and `data/houst.csv`. Read `COURSE_CONTEXT.md` from the same repository first — it explains how this course expects analytical work to be done.

Run the script and show me its actual output, verbatim. Then walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

If you cannot reach GitHub, tell me plainly and I will upload the files instead.

Refresh the visitor data and refit

PROMPT — THE LVCVA WORKBOOK

Run `lvcva_tools.py` to rebuild `data/lv_visitors.csv` from the LVCVA research page. Show me the provenance lines it prints, the last six rows of the file, and whether the newest month differs from the copy I uploaded earlier (`lv_visitors_old.csv`) — including any *revised* earlier months, which the LVCVA flags with an `r`. Then refit the seasonal-dummy model on January 2022 onward and give me the next-month forecast and a 12-month backtest. If the download fails, show the error and stop; do not fill in a month from memory.

Explain a coefficient in English

PROMPT — A COEFFICIENT, NOT A FORMULA

Here is the seasonal-dummy regression output for Las Vegas visitor volume (paste the coefficient table). Explain the coefficient on `SD10` to a hotel general manager in two sentences without using the

words “dummy”, “regression”, “coefficient” or “base”. Then tell me what the intercept means and why it is not the January forecast.

Make it argue

PROMPT — TREND LINE VERSUS NAIVE ON THE NEVADA HPI

Using `nvsthpi.csv` from 2015Q1 on, fit a trend projection and run the Durbin–Watson test on its residuals with `dw_test`. Show the output. Then make the case, using only the last eight residuals, that the naive forecast will beat the trend projection next quarter — and the case that it will not. End by saying which argument the Durbin–Watson result supports and why.

Verify

Check	What it catches
The seasonal factors sum to the number of seasons	Unnormalised factors, which bias every forecast the same direction
The first and last CMA rows are blank	A “centred” average that is not centred
The dummy for the base season is absent	A model with k dummies for k seasons, which will not estimate
The forecast’s season matches the calendar month being forecast	A factor applied to the wrong month — the commonest error in this section
The units line was read before a seasonal model was fitted	A season fitted to a seasonally adjusted series
$n \geq 15$ before a Durbin–Watson verdict is reported	A test the table cannot support

Your Turn (from the textbook)

Using the GPA series (three seasons a year: Fall, Spring, Summer): find the seasonal factors (Your Turn 7), forecast the next three semesters by multiplicative decomposition (Your Turn 8), and compute its MAD (Your Turn 9). Three seasons is odd, so the moving average is already centred and there is no second averaging step — check that the code agrees.

```
          SF   NSF
season
1      1.081 1.082
2      0.893 0.893
3      1.024 1.024
    t  season     T     S  forecast
0  10       1 3.312 1.082     3.584
1  11       2 3.414 0.894     3.051
2  12       3 3.516 1.024     3.602
MAD: 0.055
```

Laboratory 16c

Path 1 — upload the files

Download the Laboratory 16c pack — one file. Unzip it and you will have a folder holding 8 files. Drag all 8 into a new Claude conversation, wait for each to show as attached, then paste this:

PROMPT

I have uploaded 8 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.

Then run `16c_regression_seasonality.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

PROMPT

Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 16c and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.

Then run `16c_regression_seasonality.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach GitHub, **nothing is wrong with your work or with the repository.** Use Path 1.

The last submissions. For each of the five series, your final index card should now name a method, a window, the parameters, and the backtested MAD — and one sentence on which component of the series the method is built to capture. If you cannot name the component, you have picked the method by its MAD alone, and a method picked by its MAD alone is the one most likely to have been fitted to noise.